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SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

Department of Artificial Intelligence and Machine Learning

Assignment Practical No. 6

Adventure Works Sales & Profitability Dashboard

Name	Manav Goel
PRN	24070126105
Assignment	Assignment Practical No. 6
Course	Data Visualization Lab (TE7663)
Semester / Year	Sem V, Academic Year 2026–2027

PUBLISH LINK

1. Objective

The aim of this practical is to design analytical KPIs and custom DAX calculations in Power BI Desktop, validate them with DAX Studio, and present the results in an interactive dashboard. The question I set out to answer: which products, categories and regions drive Adventure Works' sales and profit between January 2020 and June 2022, and how is performance changing over time?

The work covers the six required tasks:

- Create the 10 required KPI measures.
- Create at least 20 custom DAX measures (25 in total).
- Perform time-based analysis: monthly sales, previous-period sales, sales variance and growth percentage.
- Run and validate DAX queries in DAX Studio and capture screenshots.
- Build an interactive dashboard with KPI cards, at least four visuals, and slicers for year, category and region.
- Analyze the results and state the key insights and conclusion.

2. Dataset and Data Preparation

I used the Adventure Works dataset and loaded only the tables needed for the analysis. The data runs from 1 January 2020 to 30 June 2022, so 2022 covers half a year only.

Table	Rows	What it holds
Sales (2020–2022 combined)	56,046	Every order line: date, order number, product, customer, territory, quantity
Products	293	Product names, costs and prices
Subcategories	37	Connects products to categories
Categories	4	Bikes, Components, Clothing, Accessories
Territories	10	Region, country and continent
Customers	18,148	Customer details (after cleaning)
Dates	912	Calendar from 1 Jan 2020 to 30 Jun 2022

Cleaning. The three yearly sales files were combined into a single Sales table with the Folder connector. In the Customers file I removed six non-customer rows (three rows with a key like "30---", one blank row and two export-note lines) and set CustomerKey to Whole Number. This fixed the duplicate-Null error and let the Sales–Customers relationship work.

Calculated columns. The sales files hold quantities only, so I derived sales and cost from each product's price and cost:

```
Sales Amount = Sales[OrderQuantity] * RELATED(Products[ProductPrice])
Cost Amount  = Sales[OrderQuantity] * RELATED(Products[ProductCost])
```

Model. All relationships are many-to-one with a single filter direction: Sales to Products, Customers, Territories and Dates, and Products to Subcategories to Categories. The Dates table has Year, Month Number, Month Name and YearMonth columns and is marked as the date table so time-intelligence functions work correctly.

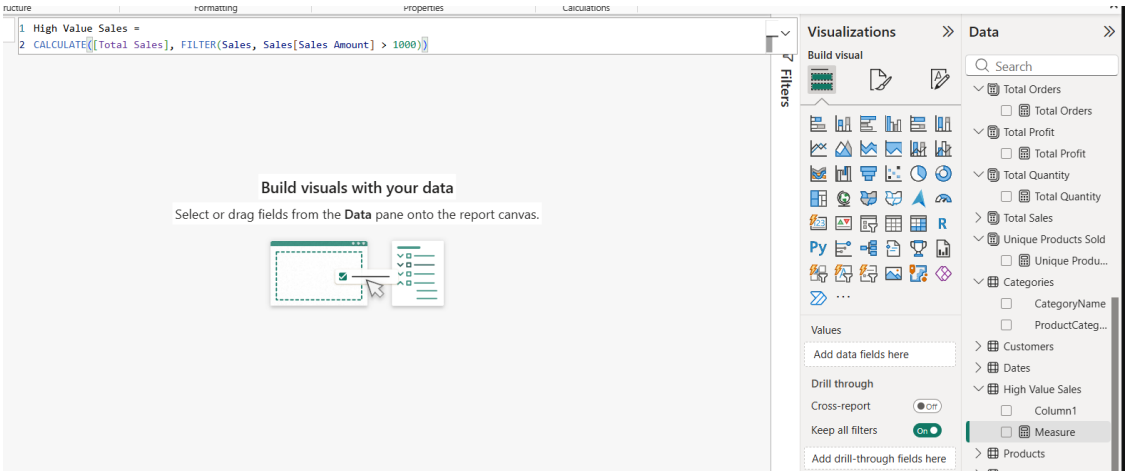
3. DAX Measures and KPI Calculations

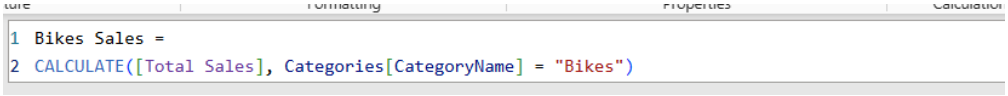
I created 25 measures in a dedicated Measures table: the 10 required KPIs, 11 additional measures that use CALCULATE, FILTER and RANKX, and 4 time-based measures.

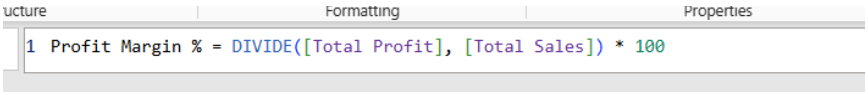
Measure	DAX	What it gives
A. Required KPI measures (10)		
Total Sales	SUM(Sales[Sales Amount])	Total sales amount
Total Cost	SUM(Sales[Cost Amount])	Total product cost
Total Profit	[Total Sales] - [Total Cost]	Sales minus cost
Total Quantity	SUM(Sales[OrderQuantity])	Units sold
Total Orders	DISTINCTCOUNT(Sales[OrderNumber])	Number of orders
Total Customers	DISTINCTCOUNT(Sales[CustomerKey])	Unique customers
Average Sales	AVERAGE(Sales[Sales Amount])	Average per order line
Average Order Value	DIVIDE([Total Sales], [Total Orders])	Sales per order

Measure	DAX	What it gives
Average Product Price	AVERAGE(Products[ProductPrice])	Average list price
Profit Margin %	DIVIDE([Total Profit], [Total Sales]) * 100	Profit as % of sales
B. Additional measures (11)		
Order Line Items	COUNT(Sales[OrderLineItem])	Number of order lines
Sales per Customer	DIVIDE([Total Sales], [Total Customers])	Revenue per customer
Profit per Order	DIVIDE([Total Profit], [Total Orders])	Profit per order
Avg Quantity per Order	DIVIDE([Total Quantity], [Total Orders])	Units per order
Unique Products Sold	DISTINCTCOUNT(Sales[ProductKey])	How many products sold
Bikes Sales	CALCULATE([Total Sales], Categories[CategoryName] = "Bikes")	Sales from Bikes only
High Value Sales	CALCULATE([Total Sales], FILTER(Sales, Sales[Sales Amount] > 1000))	Sales from lines over 1,000
Sales % of Total	DIVIDE([Total Sales], CALCULATE([Total Sales], ALLSELECTED())) * 100	Share of selected total
Product Rank	RANKX(ALL(Products[ProductName]), [Total Sales], , DESC, DENSE)	Product ranking
Region Rank	RANKX(ALL(Territories[Region]), [Total Sales], , DESC, DENSE)	Region ranking
Category Rank	RANKX(ALL(Categories[CategoryName]), [Total Sales], , DESC, DENSE)	Category ranking
C. Time-based measures (4)		
Monthly Sales	CALCULATE([Total Sales], DATESMTD(Dates[Date]))	Sales for the month
Previous Month Sales	CALCULATE([Total Sales], DATEADD(Dates[Date], -1, MONTH))	Last month's sales
Sales Variance	[Total Sales] - [Previous Month Sales]	Change from last month
Sales Growth %	DIVIDE([Sales Variance], [Previous Month Sales]) * 100	Month-on-month growth

Example screenshots of DAX measure creation

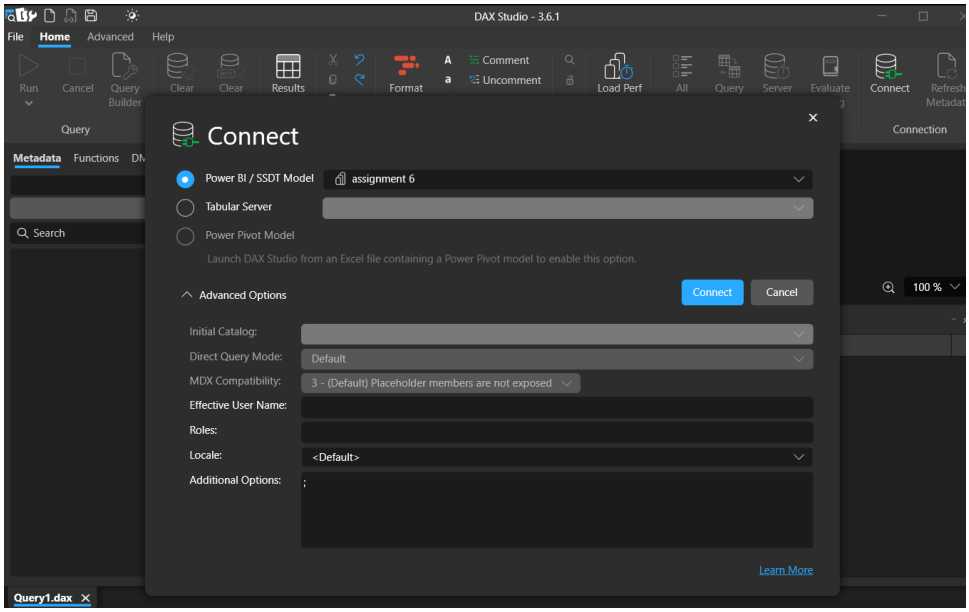






4. Validation in DAX Studio

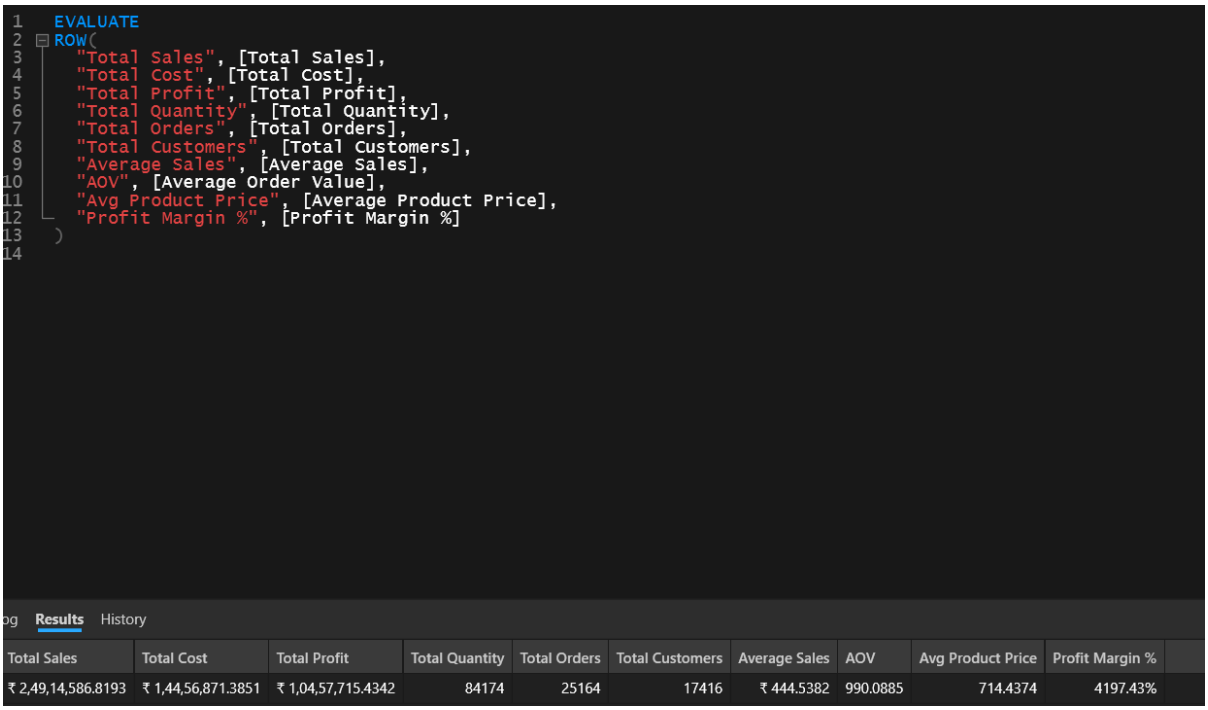
The External Tools tab did not appear in my Power BI ribbon, so I opened DAX Studio on its own, chose PBI / SSDT Model and connected to my open file. I could then run queries against the model straight away.



Query 1: KPI sanity check

This query pulls every required KPI into a single row so it can be compared with the dashboard cards.

```
EVALUATE
ROW(
  "Total Sales", [Total Sales],
  "Total Cost", [Total Cost],
  "Total Profit", [Total Profit],
  "Total Quantity", [Total Quantity],
  "Total Orders", [Total Orders],
  "Total Customers", [Total Customers],
  "Average Sales", [Average Sales],
  "AOV", [Average Order Value],
  "Avg Product Price", [Average Product Price],
  "Profit Margin %", [Profit Margin %]
)
```



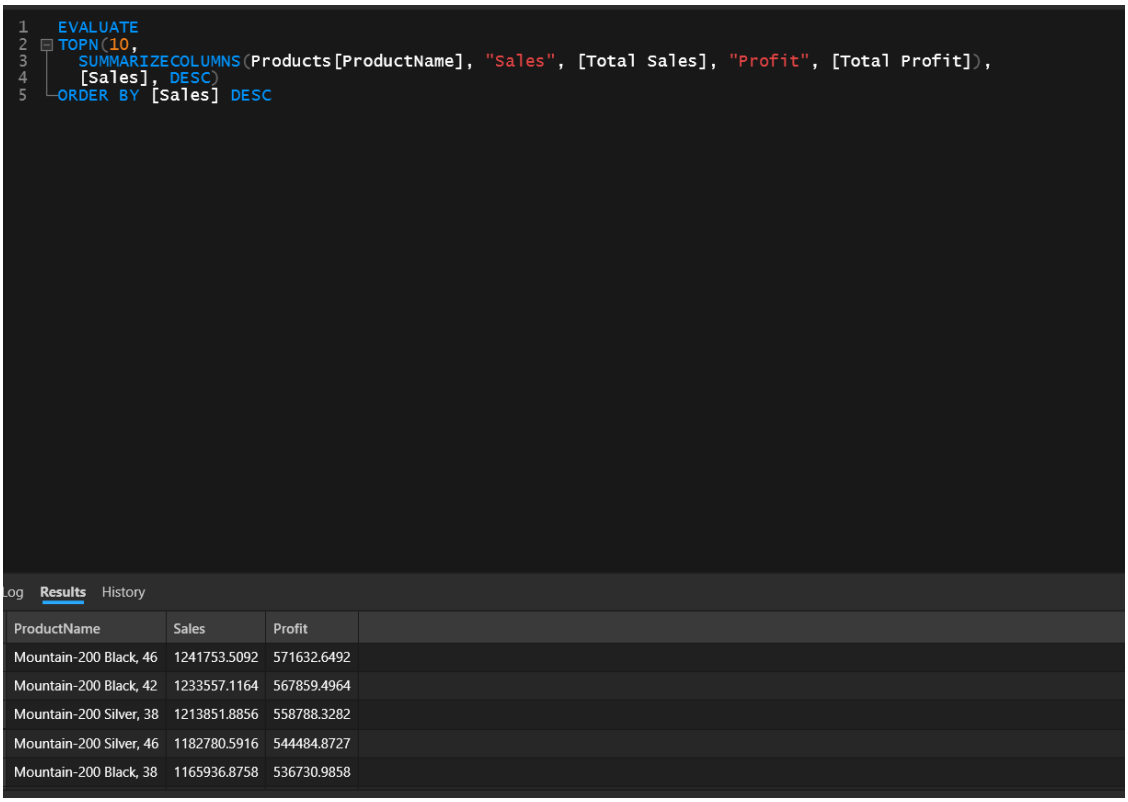
I also calculated the same totals directly from the raw CSV files, outside Power BI. Everything agreed:

Measure	Value	Matches the raw data?
Total Sales	24,914,586.82	Yes
Total Cost	14,456,871.39	Yes
Total Profit	10,457,715.43	Yes
Total Quantity	84,174	Yes
Total Orders	25,164	Yes
Total Customers	17,416	Yes
Average Sales (per line)	444.54	Yes
Average Order Value	990.09	Yes
Average Product Price	714.44	Yes

Measure	Value	Matches the raw data?
Profit Margin %	41.97	Yes

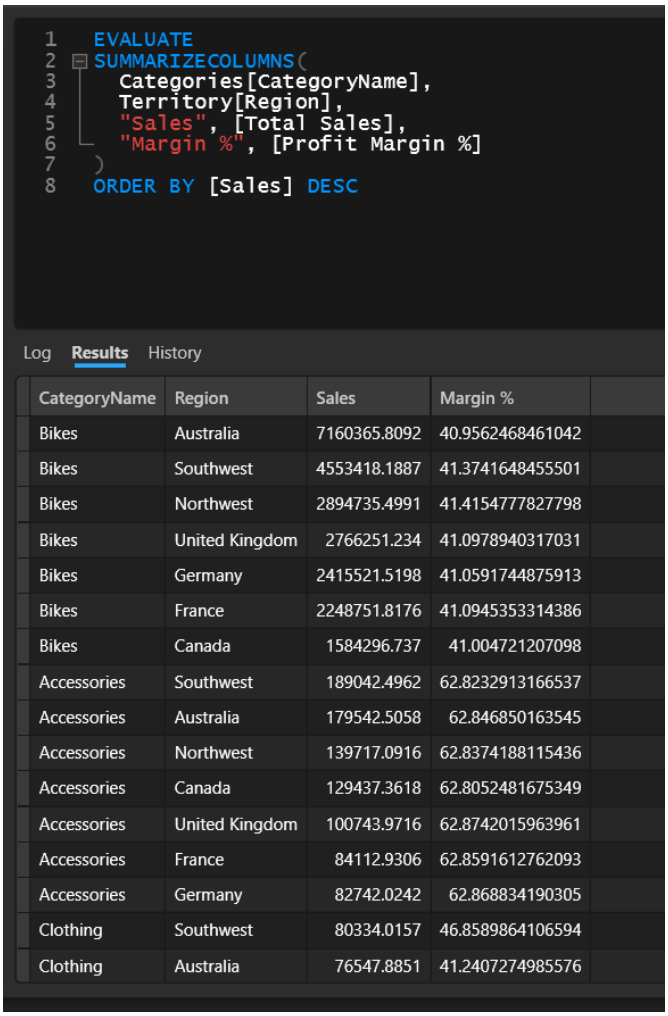
Query 2: top 10 products

```
EVALUATE
TOPN(10, SUMMARIZECOLUMNS(Products[ProductName], "Sales", [Total Sales], "Profit", [Total Profit]),
  [Sales], DESC)
ORDER BY [Sales] DESC
```



Query 3: sales and margin by category and region

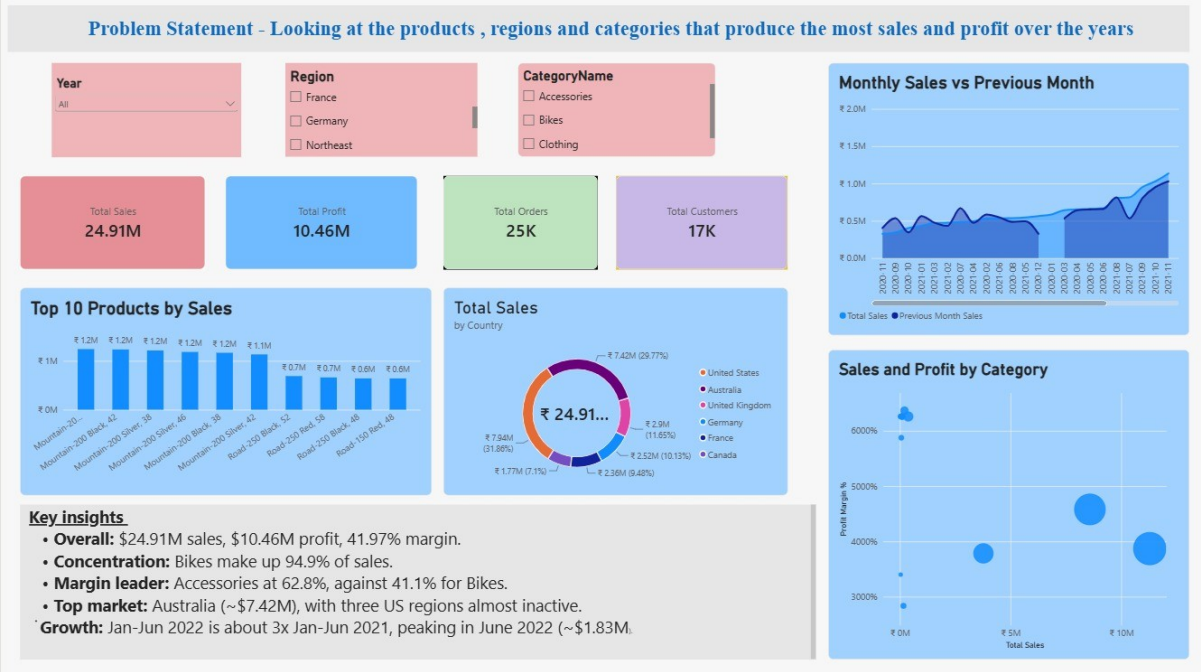
```
EVALUATE
SUMMARIZECOLUMNS(Categories[CategoryName], Territories[Region],
  "Sales", [Total Sales], "Margin %", [Profit Margin %])
ORDER BY [Sales] DESC
```



CategoryName	Region	Sales	Margin %
Bikes	Australia	7160365.8092	40.9562468461042
Bikes	Southwest	4553418.1887	41.3741648455501
Bikes	Northwest	2894735.4991	41.4154777827798
Bikes	United Kingdom	2766251.234	41.0978940317031
Bikes	Germany	2415521.5198	41.0591744875913
Bikes	France	2248751.8176	41.0945353314386
Bikes	Canada	1584296.737	41.004721207098
Accessories	Southwest	189042.4962	62.8232913166537
Accessories	Australia	179542.5058	62.846850163545
Accessories	Northwest	139717.0916	62.8374188115436
Accessories	Canada	129437.3618	62.8052481675349
Accessories	United Kingdom	100743.9716	62.8742015963961
Accessories	France	84112.9306	62.8591612762093
Accessories	Germany	82742.0242	62.868834190305
Clothing	Southwest	80334.0157	46.8589864106594
Clothing	Australia	76547.8851	41.2407274985576

5. Interactive Dashboard

The dashboard has four KPI cards (Total Sales, Total Profit, Total Orders, Total Customers), slicers for Year, Region and CategoryName, four visuals and a key-insights panel. Selecting a slicer value updates every card and chart on the page.



Visual	Fields	Why I added it
Area / line chart	X: Year-Month; Y: Total Sales, Previous Month Sales	To see the monthly trend against the previous month
Column chart	X: ProductName (Top 10); Y: Total Sales	To spot the best sellers
Donut chart	Legend: Country; Values: Total Sales	To see which countries contribute most to sales
Scatter chart	X: Total Sales; Y: Profit Margin %	To compare sales volume with profitability by category

6. Analytical Findings

- **The big picture:** about $24.91M in sales and $10.46M in profit, a 41.97% margin, from 25,164 orders placed by 17,416 customers. The average order is worth roughly $990.
- **Top-selling products:** the top of the list is all Mountain-200 bikes. Mountain-200 Black 46 leads with about $1.24M, followed by Black 42 (about $1.23M) and Silver 38 (about $1.21M).
- **Categories:** Bikes bring in about $23.64M, which is 94.9% of all sales. Accessories add roughly $0.91M and Clothing $0.37M. Components has no sales in this data.
- **Profitability:** Bikes sell the most but have the lowest margin (41.1%). Accessories earn the most per dollar sold (62.8%), and Clothing sits in the middle (44.3%). High revenue does not automatically mean the best margin.

Category	Sales ($)	Profit margin	Share of sales
Bikes	23.64M	41.1%	94.9%
Accessories	0.91M	62.8%	3.6%
Clothing	0.37M	44.3%	1.5%
Components	No sales	-	-

- **Regions:** Australia leads with about $7.42M, close to 30% of all sales. Southwest (about $4.82M), Northwest (about $3.10M) and the United Kingdom (about $2.90M) follow. Three US regions, Central, Northeast and Southeast, barely register, with under $12K each.
- **Sales trend:** sales grew from $6.40M in 2020 to $9.32M in 2021. The 2022 data covers January to June only, and it had already reached $9.19M, about three times the same six months of 2021. The monthly line keeps climbing, and June 2022 was the best month so far at about $1.83M, up around 3.3% on May.

7. Conclusion

Adventure Works' sales and profit come overwhelmingly from Bikes, led by the Mountain-200 range, and from a handful of markets, with Australia first. Accessories are the most profitable category per dollar (62.8% margin) but contribute only 3.6% of sales, so growing them is the clearest way to lift overall margin. Sales are rising steadily month on month, with the first half of 2022 already close to the full-year 2021 total. Three US regions are almost inactive, which points to untapped potential, and Components shows no sales at all, which is worth checking in the source data.

The 25 DAX measures match the raw data, and the dashboard lets these patterns be explored by year, region and category.